2023 SSMO Relay Round 1 Solutions

SMO Team

RR 1 Part 1: Compute the remainder when $2022^{2021^{2020^{\cdots}}}$ is divided by 2023.

Answer: $\boxed{2022}$

Solution: Note that

$$2021^{2020^{\cdots}} \equiv 1 \pmod 2,$$

meaning

$$2022^{2021^{2020^{\cdots}}} \equiv (-1)^{2021^{2020^{\cdots}}} \equiv (-1)^1 \equiv \boxed{2022} \pmod{2023}.$$

RR 1 Part 2: Let $T = TNYWR$. Let $a_0 = 3$, $a_1 = 1$, $a_2 = T$, and let $a_n = a_{n-1} - \frac{a_{n-3}}{8}$ for $n \geq 3$. Find

$$\sum_{i=0}^{\infty} a_i.$$

Answer: $\boxed{16176}$

Solution: Let $S = \sum_{i=0}^{\infty} a_i$, which converges since the roots $\frac{1}{2}$ and $\frac{1\pm\sqrt{5}}{4}$ of the characteristic equation $8x^3 - 8x^2 + 1 = 0$ all have absolute value less than 1. We have

$$\sum_{i=3}^{\infty} a_i = \sum_{i=2}^{\infty} a_i - \frac{1}{8}\sum_{i=0}^{\infty} a_i \implies$$

$$S - (a_0 + a_1 + a_2) = S - (a_0 + a_1) - \frac{S}{8}.$$

So,

$$\frac{S}{8} = a_2 \implies S = 8a_2 = 8T = 8 \cdot 2022 = \boxed{16176}.$$

RR 1 Part 3: Let $T = TNYWR$. Find the number of solutions to the equation

$$\sec^T(Tx) - \tan^T(Tx) = 1$$

such that $0 \leq x \leq \pi$.

Answer: $\boxed{16177}$

Solution: Firstly, note that T is even. Let $T = 2k$. The equation is equivalent to

$$\left(\sec^2(Tx)\right)^k - \left(\tan^2(Tx)\right)^k = 1.$$

Let $\alpha = \tan^2(Tx)$. We have $\sec^2(Tx) = \tan^2(Tx) + 1 = \alpha + 1$, meaning

$$(\alpha + 1)^k - \alpha^k = 1.$$

Since $\alpha \geq 0$ and $(\alpha+1)^k - \alpha^k$ is strictly increasing, we must have $\alpha = 0 \implies \tan^2(Tx) = 0 \implies \sin(Tx) = 0$. Since $0 \leq Tx \leq T\pi$, there are $T + 1 = \boxed{16177}$ solutions.

2023 SSMO Relay Round 2 Solutions

SMO Team

RR 2 Part 1: Consider the cubic polynomial $f(x) = ax^3 + bx^2 + cx + d$, where a, b, c, and d are single-digit integers, which has roots of approximately

$$x \approx -0.9518399, 0.2055095, 1.460616.$$

Compute $|f(3)|$.

Answer: $\boxed{119}$

Solution: Let the three roots be x_1, x_2, x_3. We have

$$x_1 + x_2 + x_3 \approx 0.7142856 \approx \frac{5}{7},$$

so a must be ± 7, and hence,

$$f(x) = \pm 7(x - x_1)(x - x_2)(x - x_3).$$

We compute

$$(3 - x_1)(3 - x_2)(3 - x_3) \approx (3 + 0.9518)(3 - 0.2055)(3 - 1.4606) \approx 17.00.$$

Since $f(3)$ is an integer, this gives

$$|f(3)| = 7 \cdot 17 = \boxed{119}.$$

RR 2 Part 2: Let $T = TNYWR$. Suppose that $L = \left\lfloor \sqrt{T} \right\rfloor$ points are evenly spaced around a circle. Find the number of ways to select $k \geq 3$ points such that the k-gon formed strictly contains the center of the circle.

Answer: $\boxed{708}$

Solution: We have $L = \lfloor \sqrt{119} \rfloor = 10$. The total number of ways to choose $k \geq 3$ points is

$$2^{10} - \binom{10}{0} - \binom{10}{1} - \binom{10}{2} = 968.$$

We now subtract the number of selections of $k \geq 3$ points whose convex hull does not strictly contain the center. Fix the first and last chosen points going clockwise. For the center not to be strictly contained, the clockwise arc between them can span at most 5 steps, leaving at most 4 possible points strictly between them. If a points lie strictly between them, there are $2^a - 1$ ways to choose at least one of them. Summing over $a = 1$ to 4 and multiplying by 10 for rotations yields 260 such selections. Therefore, the desired number is

$$968 - 260 = \boxed{708}.$$

RR 2 Part 3: Let $T = TNYWR$. In a committee of 2023 people, T are scientists and the rest are builders. In order to make a building, $\frac{T}{2}$ people must be chosen with at least one scientist and one builder. Find the largest nonnegative integer b such that 2^b divides the number of ways to do this.

Answer: $\boxed{0}$

Solution: The number of ways to select the committee is

$$\binom{2023}{\frac{T}{2}} - \binom{T}{\frac{T}{2}} - \binom{2023-T}{\frac{T}{2}} = \binom{2023}{354} - \binom{708}{354} - \binom{1315}{354}.$$

From Legendre's formula, we have

$$v_2\left(\binom{2023}{354}\right) = 0, \quad v_2\left(\binom{708}{354}\right) = 4, \quad \text{and} \quad v_2\left(\binom{1315}{354}\right) = 4,$$

meaning $b = \boxed{0}$.

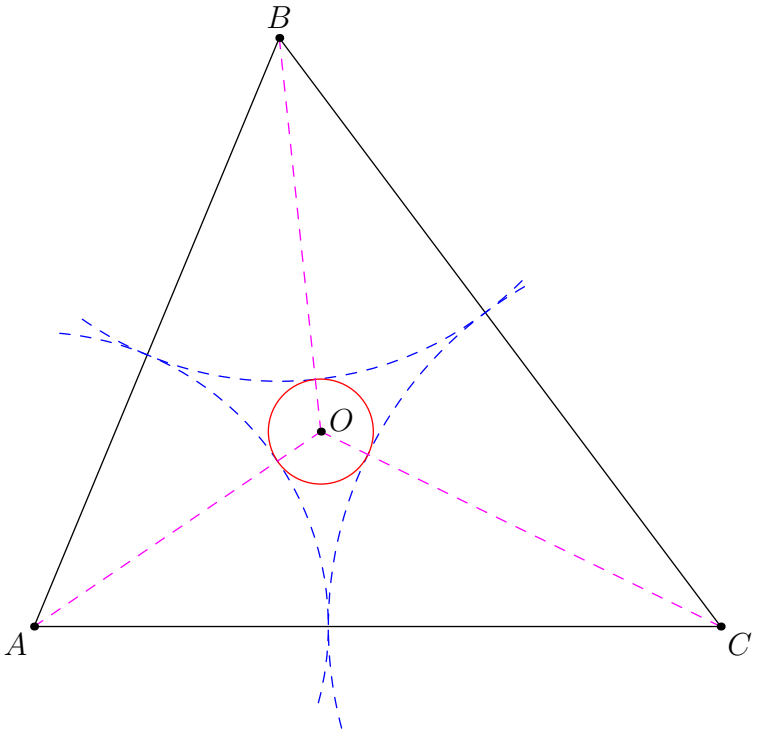
2023 SSMO Relay Round 3 Solutions

SMO Team

RR 3 Part 1: In triangle ABC with $AB = 13$, $AC = 14$, and $BC = 15$, circles ω_1, ω_2, and ω_3 are drawn, centered at A, B, and C, respectively. Each of the three circles is externally tangent to the two other circles. The radius of a circle ω such that ω is externally tangent to ω_1, ω_2, and ω_3 can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{325}$

Solution:



Let r, r_1, r_2, and r_3 be the radii of ω, ω_1, ω_2, and ω_3, respectively. Due to tangency, we have

$$r_1 + r_2 = AB, \quad r_1 + r_3 = AC, \quad \text{and} \quad r_2 + r_3 = BC,$$

which gives $r_1 = 6$, $r_2 = 7$, and $r_3 = 8$. From Descartes' Circle Theorem, we have

$$\left(\frac{1}{r} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right)^2 = 2\left(\left(\frac{1}{r}\right)^2 + \left(\frac{1}{6}\right)^2 + \left(\frac{1}{7}\right)^2 + \left(\frac{1}{8}\right)^2\right),$$

meaning

$$r = \frac{1}{\frac{1}{6} + \frac{1}{7} + \frac{1}{8} + 2\sqrt{\frac{1}{6\cdot 7} + \frac{1}{6\cdot 8} + \frac{1}{7\cdot 8}}} = \frac{168}{157} \implies 168 + 157 = \boxed{325}.$$

RR 3 Part 2: Let $T = TNYWR$. A spray painter paints all points in the coordinate plane within Euclidean distance 2 of their path. The painter walks along the perimeters of concentric squares centered at the origin, starting with the outermost square and moving inward to the innermost square:

- For each integer t from T down to 1, the painter walks along the square with vertices $(\pm t, \pm t)$.
- The spray color alternates based on t: the painter sprays blue if t is even and red if t is odd.
- If a region is painted multiple times, only the most recently applied color is visible.

The positive difference between the visible area of the blue region and the visible area of the red region can be written as $a + b\pi$, where a and b are integers. Find the remainder when $a + b$ is divided by 100.

Answer: $\boxed{20}$

Solution: Let D_t denote the total region covered by square t (along with its interior). The area of D_t is

$$[D_t] = (2t)^2 + 4(2t \cdot 2) + \pi(2^2) = 4t^2 + 16t + 4\pi.$$

Since the painter moves from the outside in, each square completely overwrites any overlap inside it. Hence, for each $t \geq 2$, the visible area added by square t is

$$[D_t] - [D_{t-1}] = \left(4t^2 + 16t + 4\pi\right) - \left(4(t-1)^2 + 16(t-1) + 4\pi\right) = 8t + 12.$$

Since $T = 325$, we can pair up the squares $(2, 3), (4, 5), \ldots, (324, 325)$. It follows that the visible red area exceeds the visible blue area by

$$\begin{aligned}[D_1] + \sum_{k=1}^{162} ((8(2k+1) + 12) - (8(2k) + 12)) &= (20 + 4\pi) + \sum_{k=1}^{162} 8 \\ &= 20 + 4\pi + 162 \cdot 8 \\ &= 1316 + 4\pi.\end{aligned}$$

So, $a + b = 1316 + 4 = 1320 \equiv \boxed{20} \pmod{100}$.

RR 3 Part 3: Let $T = TNYWR$. A town is laid out as a $T \times T$ grid of unit squares with $T + 1$ vertical lines and $T + 1$ horizontal lines, forming $2T^2 + 2T$ unit-length segments. A snowplow can start and end anywhere and must clear every segment at least once. What is the minimum number of units the snowplow must drive?

Answer: $\boxed{878}$

Solution: Note that there are $2T^2 + 2T = 840$ segments. The snowplow can drive along every segment exactly once if and only if at most 2 intersections are the endpoint of an odd number of segments. Each of the $4 \cdot 19 = 76$ non-corner intersections on the boundary is the endpoint of 3 segments, while every other intersection is the endpoint of 2 or 4. So, at least $76 - 2 = 74$ of these intersections must be the endpoint of a segment that is driven twice, meaning at least $\frac{74}{2} = 37$ segments are driven twice. If exactly 37 segments are driven twice,

then each of them joins two adjacent non-corner intersections on the same side. However, each side has 19 non-corner intersections, so at most 18 of them can be paired up this way, giving at most $4 \cdot 18 = 72 < 74$, a contradiction. So, at least 38 segments are driven twice. This is achievable by repeating 9 segments on each side, leaving out the intersections next to the same corner on two of the sides, and then repeating the 2 segments through that corner. Thus, the answer is $840 + 38 = \boxed{878}$.

2023 SSMO Relay Round 4 Solutions

SMO Team

RR 4 Part 1: Five numbers are chosen independently and uniformly at random from the set $\{1, 2, 3, 4, 5\}$, with replacement. The probability that the median of these 5 numbers is greater than 3 can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{4117}$

Solution: The median is greater than 3 exactly when at least three of the five chosen numbers are greater than 3. Each number is greater than 3 with probability $\frac{2}{5}$, independently of the others, so the desired probability is

$$\begin{aligned}&\binom{5}{3}\left(\frac{2}{5}\right)^3\left(\frac{3}{5}\right)^2+\binom{5}{4}\left(\frac{2}{5}\right)^4\left(\frac{3}{5}\right)+\binom{5}{5}\left(\frac{2}{5}\right)^5\\ =&\frac{720+240+32}{3125}=\frac{992}{3125}\implies 992+3125=\boxed{4117}.\end{aligned}$$

RR 4 Part 2: Let $T = TNYWR$. Martha starts out with the number 7 on a calculator with three buttons that multiply the current display by $\frac{1}{2}$, $\frac{2}{3}$, and $\frac{5}{6}$, respectively. She performs T rounds of operations. In each round, she chooses one of the three buttons independently and uniformly at random and presses it three times in a row. The expected value of the final value shown on her calculator can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find the remainder when $m + n$ is divided by 100.

Answer: $\boxed{70}$

Solution: Let x_i be the multiplier of the button chosen in the i-th round, so the i-th round multiplies the display by x_i^3. Since the button presses are independent, we have

$$\mathbb{E}\left(7\prod_{i=1}^{T}x_i^3\right)=7\prod_{i=1}^{T}\mathbb{E}\left(x_i^3\right).$$

Now,

$$\mathbb{E}\left(x_i^3\right)=\frac{\left(\frac{1}{2}\right)^3+\left(\frac{2}{3}\right)^3+\left(\frac{5}{6}\right)^3}{3}=\frac{\frac{27+64+125}{216}}{3}=\frac{1}{3},$$

so the expected value is $\frac{7}{3^T}$, meaning $m + n = 7 + 3^{4117}$. Since $3^{20} \equiv 1 \pmod{100}$ and $4117 \equiv 17 \pmod{20}$, we have

$$7+3^{4117}\equiv 7+3^{17}\equiv 7+3^{10}\cdot 3^7\equiv 7+49\cdot 87\equiv 7+63\equiv\boxed{70}\pmod{100}.$$

RR 4 Part 3: Let $T = TNYWR$. Suppose that $\left\lfloor\sqrt{T}\right\rfloor$ points are chosen on the sides of a triangle with area 1, where each point independently chooses one of the three sides

uniformly at random and then a location on that side uniformly at random. Given that there is at least one point on each side, the expected value of the area of the polygon formed by connecting the chosen points in counterclockwise order can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{10063}$

Solution: We have $\left\lfloor \sqrt{T} \right\rfloor = \lfloor \sqrt{70} \rfloor = 8$. Let the triangle be ABC, and suppose AB, BC, and CA contain a, b, and c points, respectively. Note that the polygon is triangle ABC with a small triangle cut off at each vertex. If the closest points to A on AB and AC are a fraction x and y of the way along their sides, then the triangle cut off at A has area $xy[ABC] = xy$. Since the closest of k random points on a side to either endpoint is an expected $\frac{1}{k+1}$ of the way along it, and the points on different sides are independent, the expected area of the polygon is

$$1 - \frac{1}{(a+1)(c+1)} - \frac{1}{(a+1)(b+1)} - \frac{1}{(b+1)(c+1)}.$$

Now, there are $\frac{8!}{a!b!c!}$ ways to distribute the points so that AB, BC, and CA contain a, b, and c points, respectively. Accounting for the ways to assign these counts to the sides, consider the following table.

Points on each side	Number of ways	Expected area
$1, 1, 6$	$3 \cdot 56 = 168$	$\frac{17}{28}$
$1, 2, 5$	$6 \cdot 168 = 1008$	$\frac{25}{36}$
$1, 3, 4$	$6 \cdot 280 = 1680$	$\frac{29}{40}$
$2, 2, 4$	$3 \cdot 420 = 1260$	$\frac{34}{45}$
$2, 3, 3$	$3 \cdot 560 = 1680$	$\frac{37}{48}$

From the Principle of Inclusion and Exclusion, there are $3^8 - 3 \cdot 2^8 + 3 = 5796$ total ways, matching the sum of the second column. So, the expected area is

$$\frac{168 \cdot \frac{17}{28} + 1008 \cdot \frac{25}{36} + 1680 \cdot \frac{29}{40} + 1260 \cdot \frac{34}{45} + 1680 \cdot \frac{37}{48}}{5796} = \frac{102 + 700 + 1218 + 952 + 1295}{5796}$$
$$= \frac{4267}{5796} \implies 4267 + 5796 = \boxed{10063}.$$

2023 SSMO Relay Round 5 Solutions

SMO Team

RR 5 Part 1: Let S_n be the set of all rational numbers of the form $0.\overline{a_1a_2a_3\ldots a_n}$, where $a_1, a_2, \ldots, a_n$ are nonzero digits. Find the value of

$$5\left(\sum_{n=1}^{\infty}\left(\sum_{a\in S_n}\frac{a}{10^n}\right)\right).$$

Answer: $\boxed{25}$

Solution: Note that the size of S_n is 9^n. For $a \in S_n$, where $a = 0.\overline{a_1a_2a_3\ldots a_n}$, the expected value of each digit a_i is

$$\frac{1+2+3+\cdots+9}{9} = 5.$$

So, the expected value of a is $0.\underbrace{\overline{55\ldots5}}_{n\text{ 5's}} = \frac{5}{9}$. So,

$$\sum_{a\in S_n}\frac{a}{10^n} = \frac{\frac{5}{9}\cdot 9^n}{10^n},$$

meaning

$$\begin{aligned}5\left(\sum_{n=1}^{\infty}\left(\sum_{a\in S_n}\frac{a}{10^n}\right)\right) &= 5\left(\sum_{n=1}^{\infty}\frac{\frac{5}{9}\cdot 9^n}{10^n}\right)\\ &= \frac{25}{9}\left(\sum_{n=1}^{\infty}\frac{9^n}{10^n}\right)\\ &= \frac{25}{9}\left(\frac{\frac{9}{10}}{1-\frac{9}{10}}\right) = \boxed{25}.\end{aligned}$$

RR 5 Part 2: Let $T = TNYWR$. Compute

$$\left(\sum_{i=1}^{T}\text{lcm}\{1,2,\ldots,i\}\right) \pmod{720}.$$

Answer: $\boxed{321}$

Solution: Let $L_i = \text{lcm}\{1,2,\ldots,i\}$. Since $720 = 2^4\cdot 3^2\cdot 5$, we have $720 \mid L_i$ for all

$i \geq 16$. For $9 \leq i \leq 15$, we have $v_2(L_i) = 3$, $v_3(L_i) = 2$, and $5 \mid L_i$, so L_i is an odd multiple of 360, meaning $L_i \equiv 360 \pmod{720}$. Since $T = 25 \geq 16$, the sum is congruent to

$$\begin{aligned} &L_1 + L_2 + \cdots + L_8 + 7 \cdot 360 \\ &\equiv 1 + 2 + 6 + 12 + 60 + 60 + 420 + 840 + 2520 \\ &\equiv 3921 \equiv \boxed{321} \pmod{720}. \end{aligned}$$

RR 5 Part 3: Let $T = TNYWR$. Distinct integers $a, b \in \{1, 2, \ldots, T\}$ are chosen uniformly at random. Consider the maximum of $ax + by$ over all reals x and y that satisfy $x^2 + y^2 = T$. Find the remainder when the expected value of the square of this maximum is divided by 100.

Answer: $\boxed{22}$

Solution: From the Cauchy-Schwarz Inequality, we have

$$(x^2 + y^2)(a^2 + b^2) \geq (ax + by)^2,$$

with equality when (x, y) is a positive multiple of (a, b). So, the square of the maximum is $T(a^2 + b^2)$, and it remains to find the expected value of $a^2 + b^2$. We have

$$\mathbb{E}(a^2 + b^2) = \mathbb{E}(a^2) + \mathbb{E}(b^2) = 2 \cdot \frac{\sum_{i=1}^{T} i^2}{T} = \frac{(T+1)(2T+1)}{3}.$$

So, the expected value of the square of the maximum is

$$T\left(\frac{(T+1)(2T+1)}{3}\right) = \frac{321 \cdot 322 \cdot 643}{3} \equiv \boxed{22} \pmod{100}.$$